

Learning from Examples

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can fit a threshold by tracing predictions and counting disagreements for every candidate.

I can apply an explicit tie rule while recording what the fitting examples leave unresolved.

I can use a selected rule on a new input and explain why fit alone does not establish future correctness.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Training examples influence how a model handles later inputs.



LEARNING TARGET Training cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit training. Choose one new labelled card length that would distinguish them and explain what remains unknown.

Learner name:

Readiness for the connected explanation

A rule predicts tall when height is at least 5 cm. What does it predict at 4, 5 and 6 cm? Why is 5 a useful boundary test?

This week's end goal

By the end, I can compare candidate models, resolve a tie explicitly and separate fit from prediction.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY A learner shown mostly striped triangles may wrongly treat stripes as the important clue for the triangle label.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

Four imaginary paper towers have heights 2, 4, 6 and 8 centimetres. Their supplied labels are short, short, tall and tall. These labels belong to this classroom task; they are not a universal definition of tall. We will fit a rule using height as its only input.

Every candidate predicts tall when height is at least its threshold and short otherwise. Compare thresholds 3, 5, 6 and 7 centimetres. The fitting procedure chooses the fewest disagreements with the four labels. If candidates tie, choose the smaller threshold. Write these choices before comparing results.

For each candidate, start its count at zero. For each example, predict with that candidate, compare with the supplied label, and add one only for a disagreement. After all four examples, record the count. Then move to the next candidate with a fresh count. This uses the conditions and repetition you have already traced.

3. See how it works

VISUAL MECHANISM

Input → **Process** → **Output**

Follow the information instead of imagining magic inside the machine.



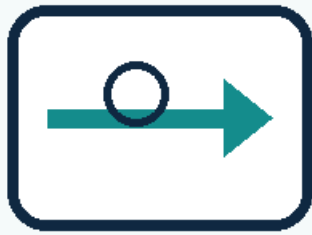
INPUT

labeled example cards



PROCESS

adjust a decision pattern from recurring
clues



OUTPUT

prediction for a new card

INPUT labeled example cards

PROCESS adjust a decision pattern from recurring clues

OUTPUT prediction for a new card

CORE IDEA Training examples influence how a model handles later inputs.

UNDER THE HOOD Supervised learning uses labeled examples to fit a mapping from represented inputs to outputs. The model does not store a human rule automatically; its fitted behavior reflects correlations, coverage, labels, and optimization choices.

Draw a fresh example with all three stages:

A closer explanation

Disagreements on the four fitting examples

Candidate · Disagreements

Candidate: Threshold 3

Disagreements: 1 of 4

Candidate: Threshold 5

Disagreements: 0 of 4

Candidate: Threshold 6

Disagreements: 0 of 4

Candidate: Threshold 7

Disagreements: 1 of 4

All candidates use the same four examples and the same zero to four scale. Thresholds 5 and 6 have zero-length bars. They tie; our stated smaller-threshold rule selects 5.

Threshold 3 incorrectly calls the height 4 tower tall. Threshold 7 incorrectly calls the height 6 tower short. Thresholds 5 and 6 agree with all four supplied labels. Check each disagreement against the individual example instead of trusting a total without its trace.

The bar chart compares disagreement counts, so a shorter bar is better on this fitting set. Zero is a measured count here, not missing data. This chart says nothing yet about the error count on new towers.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



Balanced shapes



All triangles striped



Mislabeled circle



New dotted triangle

CASE 1 — Balanced shapes Clue: colors vary within each shape. Meaning: shape clue becomes clearer.

CASE 2 — All triangles striped Clue: stripe and triangle always co-occur. Meaning: confounded clue.

CASE 3 — Mislabeled circle Clue: target is incorrect. Meaning: label noise.

CASE 4 — New dotted triangle Clue: unseen combination. Meaning: tests learned behavior.

BOUNDARY Learning from examples is not the same as memorizing every example or understanding the concept like a person.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: Reverse the tie policy to prefer the larger boundary.

Prediction: Boundary 7 is selected and predicts short at 6.

Reason: The training fit is unchanged; only the explicit choice between tied candidates changes.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





pattern



training



Word · Meaning · My example

Word: training example

Meaning: input paired with a target label

My example: _____

Word: pattern

Meaning: recurring relationship across examples

My example: _____

Word: training

Meaning: fitting behavior from examples

My example: _____

Word: new input

Meaning: case not used in fitting

My example: _____

Misconception detective

CLAIM TO REPAIR The model discovers the one true rule hidden in nature.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: Candidate model

Meaning: One possible rule being compared during fitting

Word: Fitting set

Meaning: The examples used to choose or adjust a model

Word: Disagreement count

Meaning: The number of predictions that differ from their comparison labels

Word: Tie rule

Meaning: An explicit method for choosing between equally scoring candidates

Word: Generalisation

Meaning: How well a fitted model works on new cases beyond its fitting examples

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A learner shown mostly striped triangles may wrongly treat stripes as the important clue for the triangle label.

STEP 1 — OBSERVE labeled example cards

STEP 2 — TRACE adjust a decision pattern from recurring clues

STEP 3 — CONCLUDE prediction for a new card

MODEL REASONING Early examples may support several rules. New examples eliminate some possibilities and may change the decision boundary. Record each revision as evidence that training data shapes later behavior.

DO NOT OVERCLAIM Learning from examples is not the same as memorizing every example or understanding the concept like a person.

What new evidence could change this answer?

A closer explanation

Height and supplied label · Prediction under threshold 5 · Prediction under threshold 6

Height and supplied label: 2 cm short

Prediction under threshold 5: Short

Prediction under threshold 6: Short

Height and supplied label: 4 cm short

Prediction under threshold 5: Short

Prediction under threshold 6: Short

Height and supplied label: 6 cm tall

Prediction under threshold 5: Tall

Prediction under threshold 6: Tall

Height and supplied label: 8 cm tall

Prediction under threshold 5: Tall

Prediction under threshold 6: Tall

Height and supplied label: 5 cm label not yet supplied

Prediction under threshold 5: Tall

Prediction under threshold 6: Short

The first four rows are fitting examples. The final row asks what the two tied candidates predict at a height missing from that set. Our tie rule selects 5, so the selected model predicts tall at 5. The fitting examples alone do not tell us whether that new prediction matches the task label.

A tie rule makes the selection reproducible. It does not prove that the chosen threshold is the true boundary. Record that 6 tied with 5 and that they disagree at height 5. Hiding the tie would hide a real limitation of the available evidence.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learning from Examples

A fictional classifier learns whether measured sticks are short or long.

The facts we will use

Training lengths 2,4,8,10 cm have labels short,short,long,long.

Candidate boundaries: long at least 5 or at least 7. Lowest errors wins; tie chooses smaller boundary.

AI ACADEMY / AI-1/13

Same training errors, different new prediction

	Boundary	Training errors	Prediction at 6
Candidate 5	5	0	long
Candidate 7	7	0	short

Read a cell using both its row and column labels.

Learning from Examples

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training rows

2 and 4 are short; 8 and 10 are long.



2. Candidate checks

Boundaries 5 and 7 both predict S,S,L,L: zero errors each.



3. Tie policy

Choose boundary 5 because the declared tie rule prefers the smaller value.



4. New prediction

Length 6 is predicted long by the selected rule.

CHANGE ONE THING

Reverse the tie policy to prefer the larger boundary.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Training rows

What happens and why: 2 and 4 are short; 8 and 10 are long.

Step: 2. Candidate checks

What happens and why: Boundaries 5 and 7 both predict S,S,L,L: zero errors each.

Step: 3. Tie policy

What happens and why: Choose boundary 5 because the declared tie rule prefers the smaller value.

Step: 4. New prediction

What happens and why: Length 6 is predicted long by the selected rule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

There is no training stick between 4 and 8 to distinguish these candidates. The tie policy chooses one consistently; it is not additional evidence that its prediction at 6 is true. Label-free prediction applies the selected boundary, while evaluation would require an appropriate new reference label.

What this example does not establish

Zero training errors for both rules cannot determine the correct label of every unobserved stick.

Read the labelled picture: Which new measured length would reveal disagreement between the candidates?

Change one thing: Reverse the tie policy to prefer the larger boundary. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: Balanced shapes

Evidence: colors vary within each shape

What it shows: shape clue becomes clearer

Case: All triangles striped

Evidence: stripe and triangle always co-occur

What it shows: confounded clue

Case: Mislabeled circle

Evidence: target is incorrect

What it shows: label noise

Case: New dotted triangle

Evidence: unseen combination

What it shows: tests learned behavior

Repair two tempting claims

CLAIM A The model discovers the one true rule hidden in nature.

Repair:

CLAIM B One example is enough to learn a reliable category.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

The best candidate means best among the candidates compared, using this counting rule and these recorded labels. A different candidate family, feature or label policy might give a different result. People chose those ingredients; fitting does not remove those choices.

If two records with identical input height have different labels, any deterministic rule using only that height gives them the same prediction. It must disagree with at least one label. Investigate the records and task definition instead of silently changing a label to make the model look perfect.

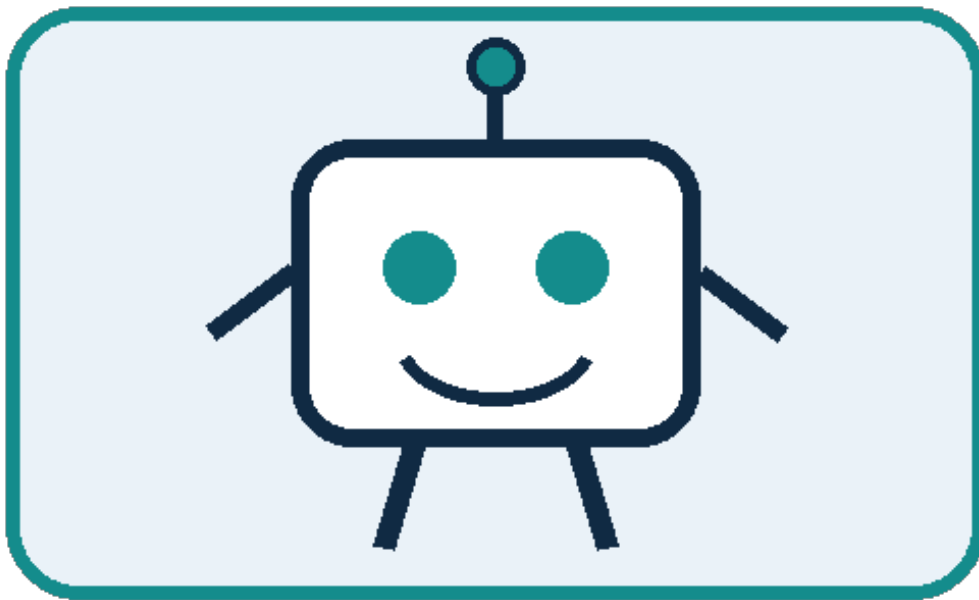
The selected threshold has zero disagreements on four fitting examples. A learner claims it will label every future tower correctly. Write a more accurate claim and name evidence that is still needed

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Training examples influence how a model handles later inputs.

Use a fresh example to explain training example.

Trace a new input → process → output example.

Find and repair the error in: One example is enough to learn a reliable category.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Suppose the task designer supplies an additional fitting example: height 5 centimetres, label short. Compare thresholds 5 and 6 again using all five fitting examples. Do not discard the original four when adding the new example.

List the prediction at the added height 5 under each candidate and update both total disagreement counts. Select between 5 and 6 under the fitting procedure. Explain why the added example is now fitting evidence rather than an independent final test

A label that helps choose the threshold participates in fitting. This is allowed, but it changes what the example can establish. We must use separate evidence to evaluate the selected model beyond the examples that shaped it. Week 15 will develop that separation in detail.

Practise with less help: Explain a tied fit

Use workbook W2 for the connected example “Fitting a pattern from examples”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Fitting a pattern from examples. Proposed explanation: “The lowest error count is added across candidates”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which new measured length would reveal disagreement between the candidates?

My explanation using a pictured step and the given facts

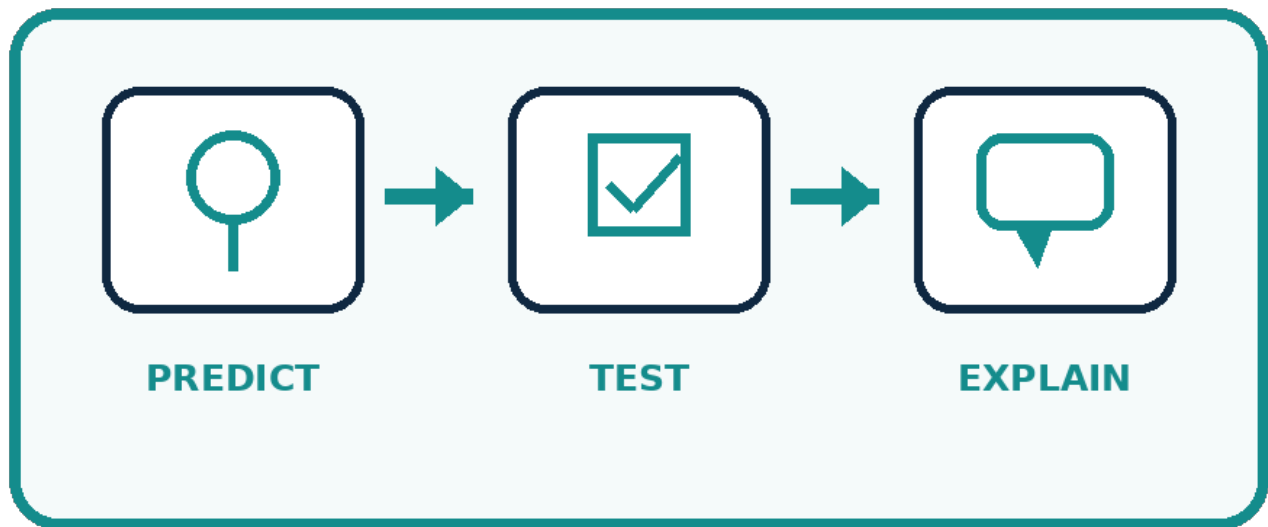
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Human Shape Model

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Act as a model trained on shape cards; record rule changes.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

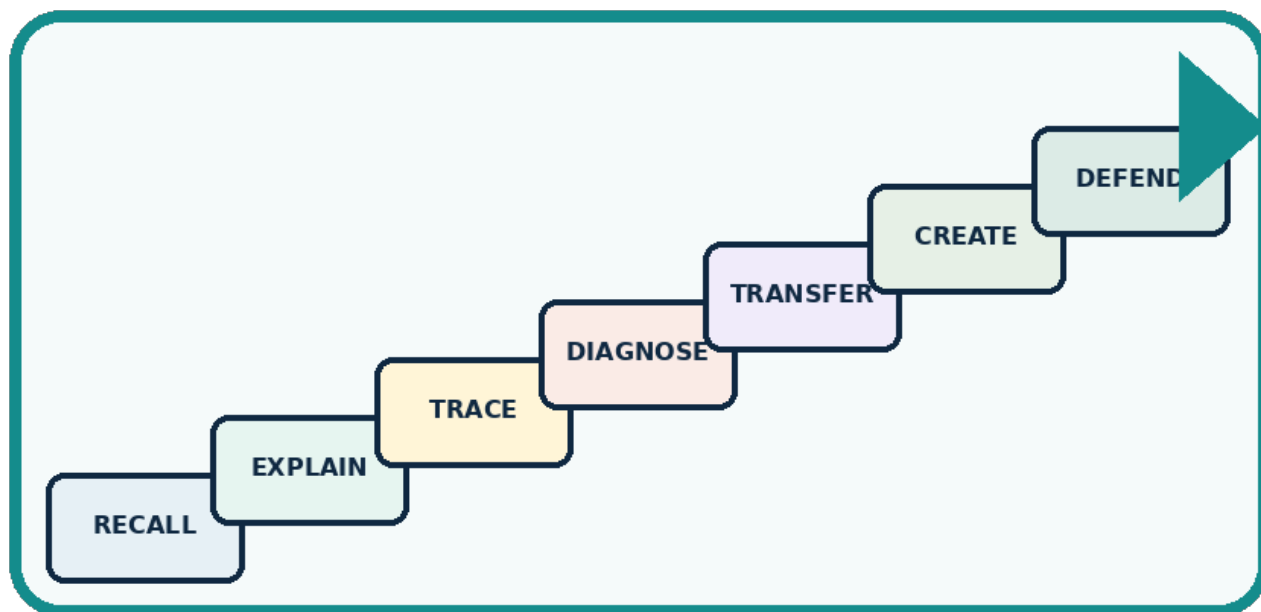
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define training example.

EXPLAIN Training examples influence how a model handles later inputs.

TRACE Show the input, process, and output.

DIAGNOSE Repair: The model discovers the one true rule hidden in nature.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

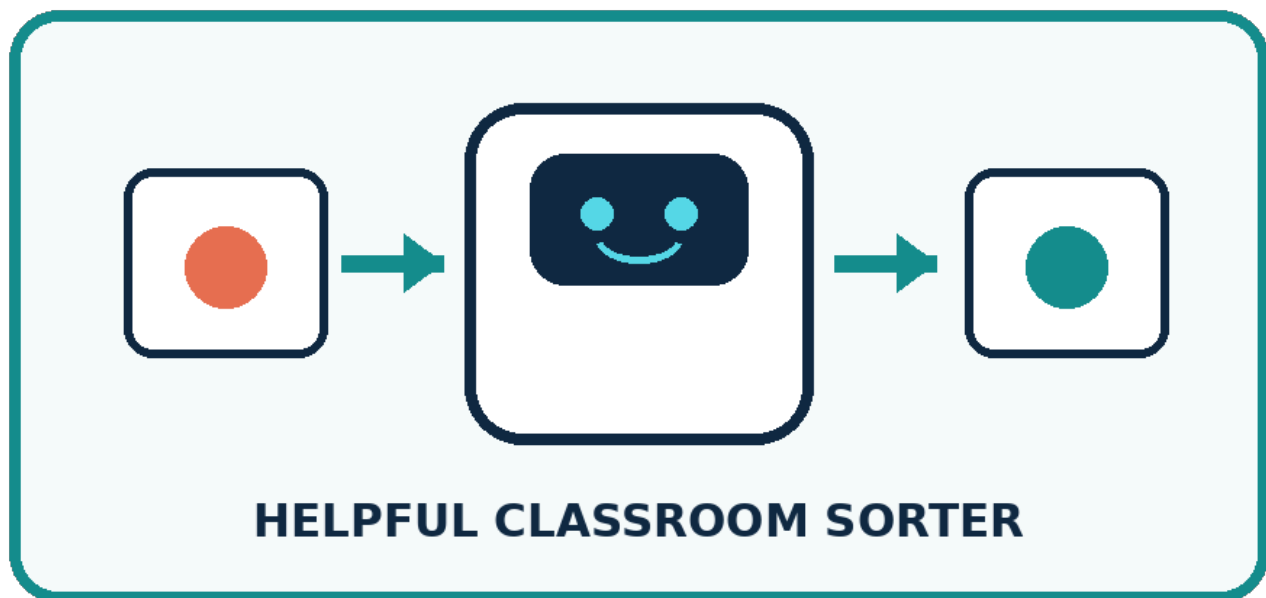
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Create the first small labeled example set for the sorter and explain which visual differences it covers.

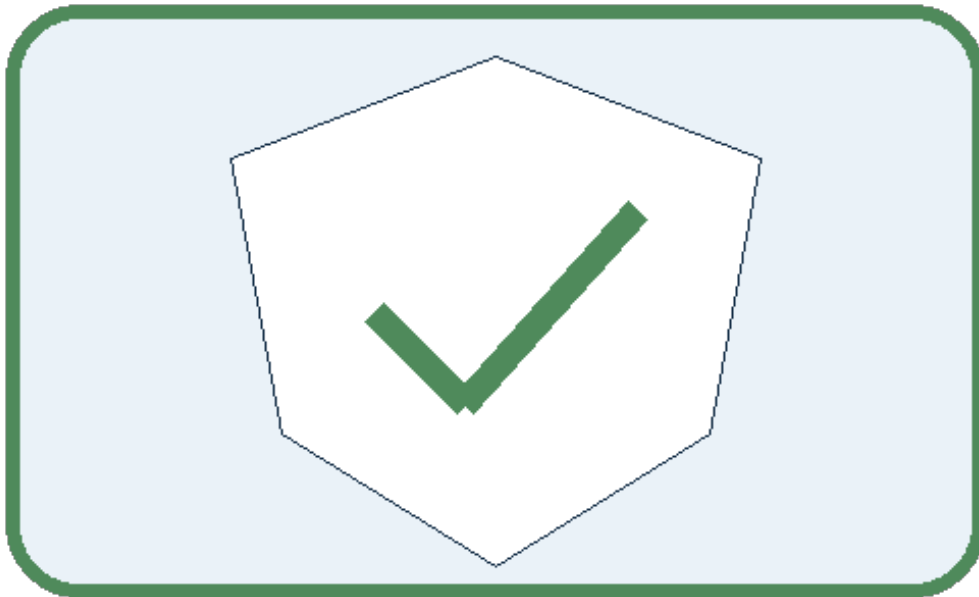
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN Learning from examples is not the same as memorizing every example or understanding the concept like a person.

My project decision:

Evidence for your project

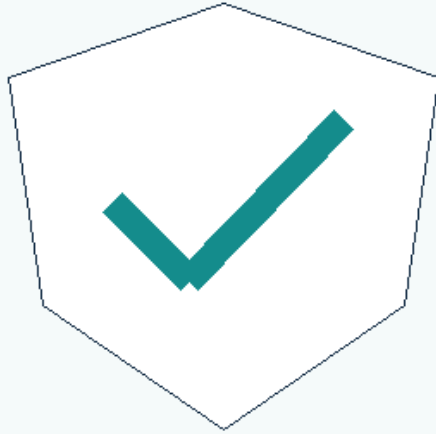
Save candidate definitions, training errors and tie policy with any sorter model version.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



INDEPENDENT PROOF

REQUIRED PROOF Train on shape cards, record how added examples change the inferred rule, and predict a new card.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK Early examples may support several rules. New examples eliminate some possibilities and may change the decision boundary. Record each revision as evidence that training data shapes later behavior.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: Build a Human Classifier.

13. Study a complete case

Week 13 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This

case extends the illustrated lesson.

The situation

Training examples label smooth leaves S and toothed leaves T. A new leaf is toothed.

The question

Predict and state the evidence limit.

Worked explanation

Predict T from the demonstrated edge pattern. One success does not establish broad reliability; use fresh labelled tests including ambiguous cases.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Examples suggest a pattern to test

Evidence or stage · Value or result

Evidence or stage: Smooth leaves

Value or result: Training label S

Evidence or stage: Toothed leaves

Value or result: Training label T

Evidence or stage: New toothed leaf

Value or result: Predict T

Evidence or stage: Fresh labelled cases

Value or result: Check the prediction rule

Training examples label smooth leaves S and toothed leaves T. A new leaf is toothed.

Predict and state the evidence limit.

Predict T from the demonstrated edge pattern. One success does not establish broad reliability; use fresh labelled tests including ambiguous cases.

14. Try a new case independently

Independent case: Fit a new distance task

Imaginary map markers have distances 1, 3, 5 and 7 units, with supplied labels near, far, far and far.

Compare candidates that predict far when distance is at least 2, 4 or 6 units. List every prediction and error count, select the lowest-error candidate, then predict at distance 2 before any comparison label is supplied. Limit your conclusion.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 13 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A learner sees three red apples and no green apples, then calls a green apple not an apple. What missing evidence could help? How would you check improvement?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create the first small labeled example set for the sorter and explain which visual differences it covers.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to

keep your practice evidence together.

Transfer practice from the original programme

Training cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit training. Choose one new labelled card length that would distinguish them and explain what remains unknown.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional classifier learns whether measured sticks are short or long.

By the end, I can · How I will show it

By the end, I can: Fit a threshold by tracing predictions and counting disagreements for every candidate

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply an explicit tie rule while recording what the fitting examples leave unresolved

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Use a selected rule on a new input and explain why fit alone does not establish future correctness

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Fit a threshold by tracing predictions and counting disagreements for every candidate

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classifier learns whether measured sticks are short or long.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Length 6 is predicted long by the selected rule.

1. Training rows
2. New prediction
3. Tie policy
4. Candidate checks

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: New prediction Length 6 is predicted long by the selected rule. There is no training stick between 4 and 8 to distinguish these candidates. The tie policy chooses one consistently; it is not additional evidence that its prediction at 6 is true. Label-free prediction applies the selected boundary, while evaluation would require an appropriate new reference label.

Practice 2: explain the deciding evidence

Changed-case practice: Reverse the tie policy to prefer the larger boundary. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Boundary 7 is selected and predicts short at 6.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Boundary 7 is selected and predicts short at 6. The training fit is unchanged; only the explicit choice between tied candidates changes.

Repair a missing building block

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose input 0 and give both predicted outputs without inventing an observed result.. Return to this week and explain what you changed.

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which input detects the difference between the two written conditions? Explain its entire run.. Return to this week and explain what you changed.

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit evidence and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.